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OBF-Psychiatric, a motor activity dataset of patients diagnosed with major depression, schizophrenia, and ADHD

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Mental health is vital to human well-being, and prevention strategies to address mental illness have a significant impact on the burden of disease and quality of life. With the recent developments in body-worn sensors, it is now possible to continuously collect data that can be used to gain insights into mental health states. This has the potential to optimize psychiatric assessment, thereby improving patient experiences and quality of life. However, access to high-quality medical data for research purposes is limited, especially regarding diagnosed psychiatric patients. To this extent, we present the *OBF-Psychiatric* dataset which comprises motor activity recordings of patients with bipolar and unipolar major depression, schizophrenia, and ADHD (attention deficit hyperactivity disorder). The dataset also contains data from a clinical sample diagnosed with various mood and anxiety disorders, as well as a healthy control group, making it suitable for building machine learning models and other analytical tools. It contains recordings from 162 individuals totalling 1565 days worth of motor activity data with a mean of 9.6 days per individual.

Background & Summary

For humans living with a psychiatric diagnosis, life is typically troublesome, as mental health problems are commonly associated with a high burden of disease and reduced quality of life, as well as occupational, financial, and personal difficulties[1]. As of today, there are no recognized biomarkers for determining and diagnosing psychiatric disorders[2,3]. Biomarkers have a diverse definition, encompassing indicators of the probability of developing a disease, medical treatment effects or side effects, and current or future pathological episodic conditions, such as psychotic episodes in schizophrenia or mood episodes in affective disorders[4]. There are no biomarkers identified by genetic studies, as there is a great overlap between the genes associated with various psychiatric diagnoses, like major depression, bipolar disorder, schizophrenia, and ADHD[5]. Although such disorders are recognized as brain disorders associated with disrupted neurobiological processes[6–8], no biomarkers have been identified by neuroimaging[2].

However, there is some optimism regarding the use of body-worn sensors to monitor biological rhythms and autonomic functions as biomarkers for humans suffering from severe psychiatric conditions[9–11]. Motor activity has been of particular interest within psychiatry[12–15], although within unipolar and bipolar disorders the focus on body-worn sensors has been more comprehensive[16–18], and, in particular, actigraphy data has shown potential as a possible future biomarker for mood state recognition[19] and even as a biomarker capable of forecasting impending mood episodes[20].

Psychiatric diagnoses are defined by the criteria of either the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders (DSM-5)[21], or the Mental, Behavioral, and Neurodevelopmental Disorders chapter of the World Health Organization's International Classification of Diseases (ICD-11)[22]. The

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two manuals are largely similar[23]. Major depression and the similar bipolar depressive state are typically characterized by a profound sense of emptiness or sadness, a diminished ability to experience joy, and decreased interest and motivation. Additional symptoms vary between individuals, but anxiety, fatigue, disrupted sleeping and eating patterns, and suicidal ideation are especially common. The lifetime risk for major depression is between 15 to 18 percent for the general population[6], and approximately 2 percent for bipolar disorder[24]. Schizophrenia is another debilitating mental disorder. While only affecting approximately one percent of the global population, symptoms of schizophrenia manifest often in early adulthood and the disorder is characterized by its chronic and relapsing nature, greatly inhibiting societal participation and the overall health of affected individuals. Schizophrenia encompasses "positive" symptoms such as hallucinations, delusions, and disorganized thoughts, "negative" symptoms, including diminished motivation, impaired emotional expression and social withdrawal, and general symptoms such as cognitive difficulties[7]. Attention-Deficit/Hyperactivity Disorder (ADHD) is a neuropsychiatric disorder that affects nearly five percent of adults, characterized by core symptoms of hyperactivity, impulsivity, and inattention. Though previously considered a childhood disorder of hyperactive boys, it is increasingly recognized in adults, often presenting with inattention and minimal gender differences. Symptoms of ADHD can significantly impact both social and occupational functioning, as well as the overall quality of life. Finally, an ADHD diagnosis frequently co-occurs with other psychiatric diagnoses, such as substance abuse, anxiety, depression and bipolar spectrum disorders[8].

Motor activity is an unobtrusive technique to continuously monitor biological rhythms and alterations in autonomic functions, and it has been hypothesized that motor activity is an objective observation of the inner psychological state expressed in behavioral patterns, in contrast to the subjective assessments ordinarily employed in psychiatric diagnostics[20]. Time series of motor activity is normally recorded with a wrist-worn actigraph device that registers gravitational acceleration in the 3-dimensional space[25]. For example, high intensity physical activities like jogging will produce higher acceleration values compared to low intensity activities like sitting. Routen, A. C. *et al.*[26] conducted an exhaustive evaluation of the device and concluded that the Actiwatch AW4 is reliable for measuring moderate and vigorous intensity activity. Motor activity patterns are an understudied topic within mental health, and most studies have been conducted in small samples of psychiatric patients[12–14]. For the development of further knowledge, larger samples of properly clinically diagnosed patients are needed. In episodic disorders, such as bipolar disorder, samples enabling comparison of mood states within individuals are in demand[19].

Given the amount of data an actigraphy device collects, the use of machine learning algorithms presents a promising method to extract insights and build predictive models[27–31]. A major obstacle for carrying out motor activity studies is data availability. Health related data, especially in psychiatric populations, is challenging to collect. Due to privacy concerns, high costs and time constraints, it is a scarce resource. This greatly limits the advancement of novel monitoring methods for mental health conditions. Thus, we present *OBF-Psychiatric*, an aggregated dataset of motor activity from humans diagnosed with major depression, schizophrenia, and ADHD, as well as healthy controls and a clinical sample diagnosed with various mood and anxiety disorders. The dataset is named in honor of the visionary and innovative professor emeritus Ole Bernt Fasmer, who designed and conducted the original actigraph studies[25,32–35]. This dataset aggregates the data from our previously published *DEPRESJON*[36], *PSYKOSE*[37], and *HYPERAKTIV*[38] datasets. These datasets have been analyzed by several groups and authors including Kumar A., *et al.*[39], Nguyen D.-K., *et al.*[40], Aminifar A., *et al.*[41], Nguyen D.-K., *et al.*[42], and Arora A., *et al.*[43], to name a few.

As opposed to previous datasets where the number of classes is limited to two (control and condition), this new dataset is suitable for multi-class machine learning algorithms and uncertainty estimation methods such as conformal prediction[44], since it contains five groups that can be tested in different settings. One could for instance train a model that classifies healthy controls, individuals diagnosed with major depression, and individuals diagnosed with schizophrenia. Although all original datasets were recorded with identical actigraph devices, they used different file conventions, column names, and different formats for dates and other variables. This new dataset standardizes all that and is homogenized into standard .csv files making it easier for researchers to read and process the recordings and simplifying feature extraction for model training and data analytics. Moreover, this dataset includes new information on migraine co-morbidity in individuals diagnosed with depression. Additionally, we include a data file with computed features that are suitable for testing machine learning models. Finally, we provide baseline experiments with the dataset by training multi-class classifiers and outlining potential future research directions using the *OBF-Psychiatric* dataset.

Methods

The anonymized *OBF-Psychiatric* dataset was originally collected for two different studies: i) a study of schizophrenia and major depression[25], and ii) a study of ADHD[33]. The Norwegian Regional Medical Research Ethics Committee West approved both studies' protocols (approval numbers: 150.01 and 251.08), written informed consent was obtained from all participants, no compensations for participating in the studies were given, and all processes were in accordance with the Helsinki Declaration of 1975[45]. The data was anonymized in accordance with the requirements of the Medical Research Ethics Committee. A specific consent for the open publication of the data was not necessarily provided by the participants; however, all participants signed an informed consent (approved by the ethics committee) before participating in the studies. In agreement with the ethics committee, at the end of the committee-approved period for retaining the study material, all patient identification materials were deleted, and the data material was anonymized in accordance with current national legislation on data anonymization[46]. The fully anonymized data can be shared and used without any restrictions under the CC-BY 4.0 license. An actigraph watch worn on the right wrist (Actiwatch, Cambridge Neurotechnology Ltd, England, model AW4) was used to monitor movement.The right wrist was chosen to make the procedure more convenient for the participants, since many had personal watches on the left wrist. This device uses a piezoelectric

	participants	average days in study	start date data collection	gender
control	32	12.5 ± 2.3	2002-10-02	female: 12, male: 20
clinical	40	7.2 ± 0.88	2009-03-12	female: 20, male: 20
depression	23	12.6 ± 2.77	2002-05-24	female: 10, male: 13
schizophrenia	22	12.95 ± 0.37	2003-09-02	female: 3, male: 19
ADHD	45	6.6 ± 1.3	2009-02-20	female: 24, male: 21

Table 1. Summary of dataset with number of participants, average days in study, start date of data collection (yyyy-mm-dd), and gender counts.

accelerometer that records the intensity, amount, and duration of the movement in gravity equivalents (g). The sampling frequency was 32Hz and magnitudes over 0.05 g are registered as an activity value in the storage unit of the device. The activity value is proportional to the intensity of the movement. The mean activity values were stored in one minute intervals. Although the presented time series data are approximately two decades old, and the 32 Hz recordings were preprocessed by the Actiwatch device and outputted as the absolute per minute, the quality and preprocessing method are comparable to methods utilized by current studies. However, the Actiwatch contained a piezoelectric omnidirectional accelerometer, while more modern actigraphs commonly contain tri-axis accelerometers. Such a difference in registering the gravitational acceleration forces has, to our knowledge, not been a problem since what we are looking for are overall patterns and not fine grained movements. For instance, Murray, G., *et al.*[47] mention that for clinical applications an epoch length of 30 seconds or 1 minute is recommended.

The dataset was collected from 23 out- and inpatients diagnosed with either unipolar or bipolar depression, from 22 long-term inpatients diagnosed with schizophrenia, and from 45 outpatients diagnosed with ADHD. All of the psychiatric wards were open. The dataset also contains data from 32 healthy controls, consisting of 23 hospital employees, 5 students, and 4 general practitioner patients without any psychiatric history, as well as a clinical sample of 40 outpatients, diagnosed with various mood or anxiety disorders. In total, the recordings account for a total of 1, 565 days of motor activity data. Table 1 presents a summary of the number of participants per group, average days in the study, start date of the data collection, and gender counts.

All patients were diagnosed by a senior psychiatrist using a semi-structured diagnostic interview based on DSM-IV criteria[48] in the depression and schizophrenia samples and DSM-IV and ICD-10 in the ADHD and clinical samples[49]. Note that the original datasets were collected around the turn of the millennium, presenting a possible limitation for the current dataset as the diagnostic manuals have been updated to newer versions since the time of collection. While there have been some modifications in the diagnostic criteria for ADHD[50], the modifications are more subtle for schizophrenia and major depression, and the foundations of the diagnostic criteria are retained in the updated versions[51,52].

Among the five inpatients and 18 outpatients diagnosed with major depression[36], the severity of the ongoing depression was rated by a clinician using the Montgomery-Asberg Depression Rating Scale (MADRS)[53] at the beginning and conclusion of the motor-activity recordings. The MADRS scale is a clinician-rated severity scale for depression. Ten items are scored, based on observation and conversation with the patient. The sum score ($0 - 60$) represents the severity of the depression. Scores below 10 are classified as absence of depressive symptoms[54] and scores above 30 indicate a severe depressive state[55]. Melancholic features of a depressive episode often include psychomotor retardation[56] and were specified as defined by DSM-IV criteria[48]. Comorbid migraine, which is prevalent in individuals with a history of depression[57], was diagnosed using the International Headache Society criteria[58].

The schizophrenia part of the dataset contains categorical information on the presence of the paranoid subtype based on DSM-IV criteria. The present symptomatic state of the patients were rated on the Brief Psychiatric Rating Scale (BPRS)[59], a frequently used rating scale for measuring the overall psychopathology of individuals diagnosed with schizophrenia. The BPRS consists of 18 items rated from 1 to 7, and higher sum scores indicate a more severe condition[60]. Seventeen of the patients were rated on the BPRS scale[37]. All of the schizophrenia diagnosed individuals used medication, specified in the dataset by psychopharmaceutical class. Comorbid migraine is also specified[58]. Generally, the first-generation antipsychotics from the 1950s were dopamine receptor antagonists and are today mainly classified as typical antipsychotics. Clozapine, the first atypical antipsychotic, changed the game when introduced in the 1970s[61]. The atypical drugs that arrived afterward differed from the typical antipsychotics in their mechanisms of action and side effect profiles[62]. Clozapine is currently considered the gold standard in treating schizophrenia[61] and has shown effects on motor activity in the sample previously[25].

The ADHD part of the dataset included consecutive participants who were referred to an outpatient clinic for diagnostic evaluation. The sample contains categorical information pertaining to ADHD diagnosis and the presence of the inattentive subtype (named ADD), as well as psychiatric comorbidity: BIPOLAR, UNIPOLAR, ANXIETY, SUBSTANCE (alcohol or drug addictions) and OTHER (the presence of additional psychiatric diagnoses). Diagnosis was set by psychiatrists based on the Mini-International Neuropsychiatric Interview (MINI Plus, version 5.0.0)[49]. The self-rated Cyclothymic Temperament scale (CT)[63] and the Mood Disorder Questionnaire (MDQ-POS)[64] were used to evaluate emotional instability and symptoms associated with bipolar spectrum disorders. CT is originally a 21-item subscale within the 110-item TEMPS-A questionnaire, rating various affective temperament dimensions[65]. The rate of psychiatric comorbidities in this sample seems high but is in accordance with the findings of a recent systematic review on psychiatric comorbidities in adult ADHD, encompassing both general and psychiatric populations[66]. The ADHD sample was also assessed with rating scales reporting sum scores, where higher scores generally indicate increased severity of symptoms. The

```
timestamp,date,activity
2003-05-07 12:00:00,2003-05-07,0
2003-05-07 12:01:00,2003-05-07,143
2003-05-07 12:02:00,2003-05-07,0
2003-05-07 12:03:00,2003-05-07,20
2003-05-07 12:04:00,2003-05-07,166
2003-05-07 12:05:00,2003-05-07,160
2003-05-07 12:06:00,2003-05-07,17
2003-05-07 12:07:00,2003-05-07,646
2003-05-07 12:08:00,2003-05-07,978
2003-05-07 12:09:00,2003-05-07,306
```

Fig. 1 First ten rows of actigraphy data for one of the patients diagnosed with depression.

18-item adult ADHD self-report scale (ASRS) evaluates current symptoms of ADHD, and outputs a sum score between 0 and 72[67]. The 25-item Wender Utah Rating Scale for ADHD (WURS) is a questionnaire that assesses the presence and severity of ADHD symptoms during childhood. The scale outputs a sum score between 0 and 100[68]. The Hospital Anxiety and Depression Scale (HADS) evaluates the severity of the present state of anxiety and depression. Seven items rate depression (HADS-D) and seven items rate anxiety (HADS-A), both with maximum scores of 21. Scores below eight on each scale indicates absence of symptoms[69]. MADRS was also used to assess depressive symptoms and is described above. Finally, medication use was categorically reported by psychopharmaceutical class.

Data Records

The *OBF-Psychiatric* dataset can be downloaded from the zenodo.org repository[70] (https://zenodo.org/records/13754984). The original data files were saved in a data format that requires specialized or commercial software to read them. With the aim of making the data portable and easier to read, we homogenized all files into a comma separated values (.csv) format. The root folder contains five subfolders. Each one contains the records for the corresponding group (adhd, clinical, control, depression, schizophrenia). For each subject, we include a *.csv* file with the actigraph data collected over time. Figure 1 shows an example of the first ten rows for one of the individuals diagnosed with depression. The columns are: timestamp (intervals of one minute), date (date of measurement) and activity (activity measurement from the device).Missing data are marked with 'NA', and for the sum scores of clinical assessments, 'NA' implies that the assessment has not been completed for this participant.

In addition, we also include information files (**-info.csv*) in the root directory where * can be *adhd*, *clinical*, *control*, *depression*, and *schizophrenia*. These files are described below.

control-info.csv. The information file for healthy controls has four columns: number (subject identifier), days (number of days in the study), acc_time (HH:MM time when the recordings started), gender (1:female, 2:male), and age (age range). The age was grouped into ranges to maintain privacy.

depression-info.csv. The information file for major depression diagnosed patients contains the following columns: number (subject identifier), days (number of days in the study), acc_time (HH:MM time when the recordings started), gender (1:female, 2:male), age (age range), afftype (1: bipolar II, 2: unipolar depressive, 3: bipolar I), melanch (1: melancholia, 2: no melancholia), migraine (1: migraine, 0: no migraine), inpatient (1: inpatient, 2: outpatient), marriage (1: married or cohabiting, 2: single), work (1: working or studying, 2: unemployed/sick leave/pension), madrs1 (MADRS score at the beginning) and madrs2 (MADRS score at the end).

schizophrenia-info.csv. The information file for patients with schizophrenia contains the following columns: number (subject identifier), gender (1:female, 2:male), age (age range), days (number of days the patient wore the device), acc_time (HH:MM time when the recordings started), schtype (paranoid, non-paranoid), migraine (migraine or no migraine), bprs (BPRS sum score), cloz (1: the patient used clozapine as antipsychotic medication, 2: did not use), antipsych (1: typical antipsychotic monotherapy, 2: atypical antipsychotic medication, clozapine included), and moodst (did the patient use mood stabilizing medications, 1: Yes, 2: No).

adhd-info.csv. The information file for patients with ADHD contains the following columns: number (subject identifier), gender (1:female, 2:male), age (age range), acc_time (HH:MM time when the recordings started), days (number of recorded days for each participant), adhd (1:diagnosed with ADHD, 0:no ADHD), add (1:presence of the inattentive subtype of ADHD), bipolar (1:presence of bipolar disorder), unipolar (1:presence of unipolar depression), anxiety (1:presence of anxiety disorder), substance (1:potential drug or alcohol addictions), other (1: presence of additional psychiatric disorders), ct (1: presence of cyclothymic temperament), and mdq_pos (1: presence of Bipolar Spectrum Disorder). The sum score scales were reported as: asrs (ASRS score), wurs (WURS score), hads_a (HADS anxiety score), hads_d (HADS depression score) and madrs (MADRS score). Prescribed medications were reported as med (1: medications prescribed) and med_*, where * can be antidepressant, mood stabilizers, antipsychotics, benzodiazepines, opioids, stimulants, and sleep aid.

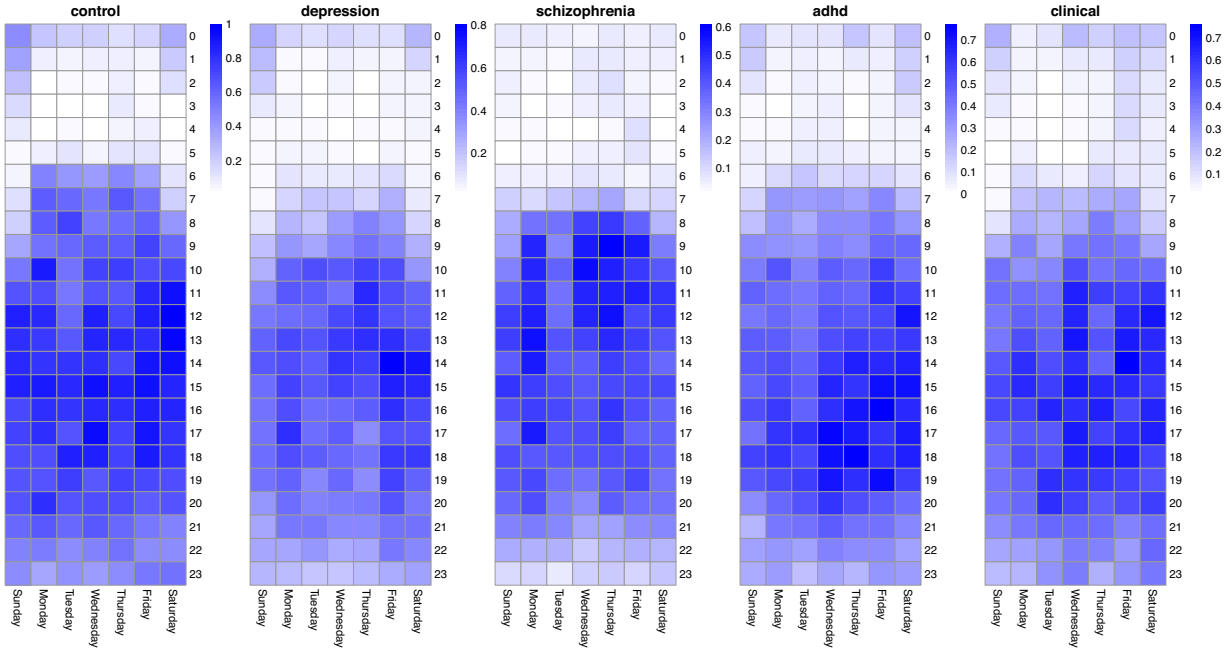

Fig. 2 Motor activity heatmaps. Each cell represents the average motor activity value across all the individuals.

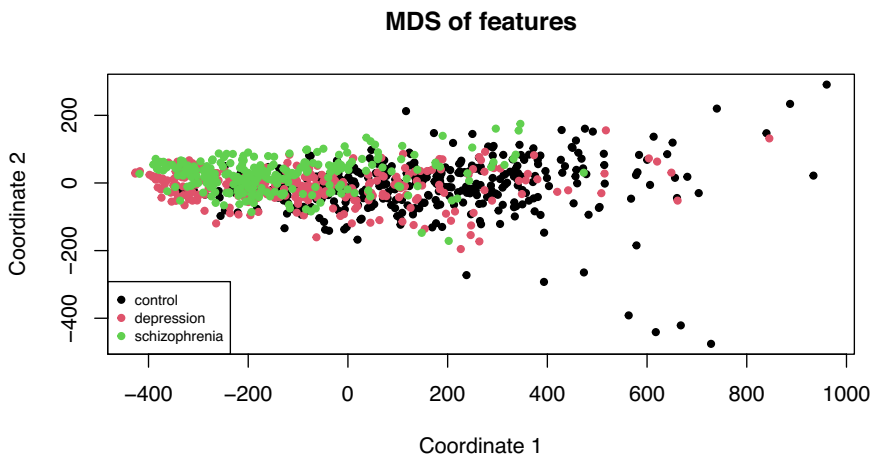

Fig. 3 Projection of the features into two dimensions using multidimensional scaling (MDS).

clinical-info.csv. The information file for the individuals diagnosed with various mood and anxiety disorders (clinical sample) contains the same columns as *adhd-info.csv*.

features.csv. We also include a file named *features.csv*. This file contains basic extracted features (from the control, depression, and schizophrenia groups) that can be used to test machine learning algorithms. Six features were extracted including the mean, standard deviation, percent of zeros, median, 25th quantile, and 75th quantile. The percent of zeros refers to the percent of readings with an activity level of 0 during the day. These features were extracted per day. That is, take the activity column for the given day and compute the mean, standard deviation, etc. Only days with complete data were included, that is, days for which there were 24*60 = 1440 activity readings. The first column refers to the subject id, the following six columns represent the features, and the last column is the type (control, depression, schizophrenia). This file is the one we used to perform the baseline experiments (see the Technical Validation section).

Technical Validation

To evaluate the technical quality and the suitability of the dataset to perform data analyses, we generated visualizations and trained several multi-class classifiers. First, we plot heatmaps for each of the five groups. The x axis represents the day of the week and the y axis the hour of the day. Each cell represents the average registered activity across all users for that particular hour/day. The values were normalized across groups so the plots are comparable. Figure 2 shows the resulting heatmaps. For the five groups, most activity occurs between 6:00 and 24:00, with some visual differences between the groups.

	Precision	Recall	F1 Score	ROC Area	Class
Random Forest	0.709	0.76	0.734	0.858	control
	0.582	0.519	0.548	0.753	depression
	0.714	0.722	0.718	0.895	schizophrenia
	0.673	0.677	0.674	**0.838**	weighted average
IBk (k=3)	0.661	0.781	0.716	0.809	control
	0.556	0.504	0.528	0.710	depression
	0.701	0.589	0.640	0.849	schizophrenia
	0.641	0.642	0.638	0.791	weighted average
Logistic Model Tree	0.707	0.790	0.746	0.850	control
	0.606	0.493	0.543	0.761	depression
	0.711	0.730	0.720	0.872	schizophrenia
	0.678	**0.683**	**0.678**	0.830	weighted average
Multilayer Perceptron	0.736	0.716	0.726	0.856	control
	0.528	0.567	0.547	0.726	depression
	0.684	0.658	0.671	0.864	schizophrenia
	0.658	0.654	0.656	0.820	weighted average

Table 2. Machine Learning classification results. Bold numbers indicate the best performing model per metric (for weighted average).

To validate the suitability of the data to train machine learning models, we extracted basic features from the activity values for each subject on a per-day basis. These features are included in the *features.csv* file. To visualize to what extent these basic features can be used to characterise each group, we used multidimensional scaling (MDS) to project the data into two dimensions. The plot was generated using the built-in *cmdscale* function in R version 4.3.2. (see the "Code availability" section for directions on obtaining the plotting script). The *cmdscale* function is based on principal coordinates analysis[71]. Figure 3 shows the resulting plot. We can see that even though the features are very basic, they still capture information that can be used to separate the three groups to some extent. Researchers using this dataset can test more complex features that may achieve better separation boundaries. For example, by extracting features during day and night hours.

Using these features, we trained four classifiers: Random Forest, K-Nearest Neighbors (IBk in Weka), Logistic Model Tree (LMT in Weka), and Multilayer Perceptron. The classifiers were used to predict the target class (control, depression, schizophrenia). The purpose of this comparative evaluation is to create a baseline for future researchers using this dataset. We conducted the experiments using a 10-fold cross validation. We trained the models using the Weka Machine Learning Software tool[72] version 3.8.6. Before training the models, we removed the *user* variable within Weka and the default settings for each classifier were used except for IBk where the number of neighbors (KNN) parameter was set to 3. We reported the performance with metrics such as precision, recall, F1-Score, and ROC area which are defined below in terms of TP (true positives), TN (true negatives), FP (false positives) and FN (false negatives).

Precision. Precision is the ratio of correctly predicted positive points to total predicted positives. The formula for precision is:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{1}$$

Recall. Recall, measures the proportion of actual positives that are correctly identified as such. It focuses on the model's ability to identify positive cases. The formula for recall is:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

F1-Score. The F1-Score is the harmonic mean of precision and recall. It is a balance between these two metrics. It is useful when the class distribution is imbalanced.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

ROC area. The ROC area measures the area underneath the receiver operating characteristic (ROC) curve. This curve shows the performance of a model at all classification thresholds based on the True Positive Rate (TPR) and False Positive Rate (FPR). It ranges from 0 to 1. The TPR (same as recall) and FPR are computed as:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{4}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \tag{5}$$

Table 2 shows the performance results for each individual class and the weighted average. In this case, Logistic Model Tree achieved the best results. These results show reasonable performance considering that the extracted features were very basic. These experiments serve as a baseline to encourage researchers to test new sets of features and methods on this dataset.

Limitations

Even though the dataset contains a lot of actigraphy data, the sample size (number of participants) is not big enough to generalize the results to other settings. We also want to note that the device was worn on the right hand but information about the dominant hand of each participant was not registered. This could be a potential confounder since a meta-analysis showed that there is evidence of elevated levels of atypical (left-, non-right-, or mixed-) handedness between individuals with ADHD and controls[73]. Comorbidities can also be relevant for the analyses. For example, the migraine rate in the group of persons with depression is 73%. In the ADHD group, 40% of the patients are diagnosed with anxiety which is within the rate of 25% − 50% mentioned by Koyuncu, A., *et al.*[74]. While Salvi, V., *et al.*[75] mention comorbidities of ADHD with bipolar disorder with rates as high as 20%, in this dataset the rate is even higher (35%). Although information about comorbidities is included in the dataset, it should be used with caution since the sample rates may not be representative of the general population. Currently, only actigraphy data is provided which can also limit the possibility to capture more specific behavior patterns. More advanced actigraphy devices capable of collecting more information (heart rate, oxygen saturation, temperature, etc.) are currently being developed and employed. Still, we believe that analyzing motor activity is a good starting point towards the development of more advanced tools and analyses to understand the studied behaviors.

Code availability

The code used to extract the *features.csv* file is available here: https://github.com/enriquegit/OBF-Psychiatric-dataset. It consists of scripts written in the R programming language. The R version used to run the scripts was 4.3.2.

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Author contributions

E.G.C. and P.J. integrated, merged, and cleaned the datasets. M.A.R. set up the database repository. E.G.C. and M.A.R. generated the plots and visualizations. E.G.C. conducted the baseline experiments. P.J. and A.S. wrote and reviewed the medical sections of the manuscript. S.H. worked on the tables and latex code. G.O. reviewed the writing and checked the conformity with the guidelines. P.H. reviewed the manuscript and provided insightful recommendations. K.J.O. has been a co-worker on all previous datasets and was involved in conceiving them. J.O.B. collected the psychosis data and most of the healthy controls. K.M. and W. F. collected the ADHD data and the clinical control samples. All authors analyzed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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